

Problem Set 3

Due Friday, October 16

Instructions. Turn in at least 30 points' worth of problems.

Problem 1. (7 points) Show that at least one of the sequences $\{2^{2^n} + 1\}_{n \geq 0}$ and $\{6^{2^n} + 1\}_{n \geq 0}$ contains infinitely many composite terms.

Problem 2. *equivalent forms of the prime number theorem* (7 points) Recall that $\psi(x) := \sum_{p^k \leq x} \log p$. Prove that if $\psi(x) \sim x$ as $x \rightarrow \infty$, then $\pi(x) \sim x / \log x$, and vice versa.

One way to solve this problem is to prove both asymptotic relations hold. Needless to say, that is not what is intended here...

Problem 3. *equivalent error terms in the prime number theorem* (10 points) Define $\theta(x) := \sum_{p \leq x} \log p$. Prove that the following are equivalent:

- (a) $\psi(x) = x + O(x/(\log x)^{100})$ for $x \geq 3$,
- (b) $\theta(x) = x + O(x/(\log x)^{100})$ for $x \geq 3$,
- (c) $\pi(x) = \text{Li}(x) + O(x/(\log x)^{101})$ for $x \geq 3$.

Here $\text{Li}(x) := \int_2^x dt / \log t$.

Partial summation is your friend.

Problem 4. (10 points)

- (a) Show that there is a constant $C > 0$ such that, for all $x \geq 3$,

$$\frac{1}{x} \sum_{n \leq x} \omega(n) = \log \log x + C + O(1/\log x).$$

Hence, positive integers $n \leq x$ have about $\log \log x$ distinct prime factors, on average.

- (b) Show that for every $\epsilon > 0$ and all large enough n ,

$$\omega(n) < (1 + \epsilon) \log n / \log \log n.$$

Hint. How does $\omega(n)!$ compare in size with n ?

- (c) Assume the Prime Number Theorem, i.e., that $\pi(x) \sim x / \log x$ as $x \rightarrow \infty$. Show that for every $\epsilon > 0$, there are infinitely many positive integers n for which

$$\omega(n) > (1 - \epsilon) \log n / \log \log n.$$

You may assume the result of Problem 2 even if you have not solved it.

Problem 5. (10 points) It is an open problem to show there is a fixed $k \in \mathbf{Z}^+$ for which the polynomial $n^2 + k$ assumes infinitely many prime values (as n ranges through the positive integers). Prove that for every B , there is a $k \in \mathbf{Z}^+$ for which $n^2 + k$ assumes at least B prime values.

Put the correct pigeons in the correct holes.

Problem 6. (7 points) Let k be an integer larger than 1. Show that there is at least one natural number n for which $n/\pi(n) = k$.

Look for a proof using that $\pi(x)/x \rightarrow 0$ as $x \rightarrow \infty$.

Problem 7. *integrals play nice with asymptotic relations* (8 points)

(a) Let f, g be positive-valued functions defined for $x \geq 2$. Suppose that

$$\int_2^x f(t) dt \rightarrow \infty, \quad \text{as } x \rightarrow \infty.$$

Give a careful proof that if $f(x) \sim g(x)$ as $x \rightarrow \infty$, then

$$\int_2^x f(t) dt \sim \int_2^x g(t) dt \quad \text{as } x \rightarrow \infty.$$

Do not appeal to L'Hôpital's rule! Get your hands dirty and bring in the epsilons.

(b) Give a counterexample to (a) if the hypothesis that $\int_2^x f(t) dt \rightarrow \infty$ is removed.

You should convince yourself that this justifies the replacement of $\pi(t)$ with $tW(t)$ in the heuristic for the PNT presented in class.

Problem 8. (12 points) Let $\mathcal{A}$ be a set of positive integers such that, for some positive constants c and K , and some threshold x_0 , we have

$$\left| \sum_{\substack{a \in \mathcal{A} \\ a \leq x}} \frac{\log a}{a} - c \log x \right| \leq K \quad \text{for all } x > x_0.$$

Prove that there are positive constants γ_1, γ_2 and a threshold x_1 such that

$$\gamma_1 \frac{x}{\log x} \leq \sum_{\substack{a \in \mathcal{A} \\ a \leq x}} 1 \leq \gamma_2 \frac{x}{\log x} \quad \text{for all } x > x_1.$$

Problem 9. (10 points)

(a) Show that for every nonnegative integer N ,

$$\text{lcm}[1, 2, \dots, 2N+1] \int_0^1 t^N (1-t)^N dt \in \mathbf{Z}^+.$$

(b) The integral in (a) is $< 4^{-N}$. Hence, $\text{lcm}[1, 2, \dots, 2N+1] > 4^N$ for every positive integer N .

(c) Show that $\exp \psi(N) = \text{lcm}[1, 2, \dots, N]$ for every positive integer N . Deduce that $\psi(2N+1) > N \log 4$ for every positive integer N .

From (c) it is easy to obtain another proof of the estimate $\psi(x) \geq x \log 2 + O(1)$, shown in class by a very different method.

Problem 10. (5 points) Evaluate the integral appearing in Problem 9(a) in closed form.

Using this evaluation and Stirling's formula, one can show that $\text{lcm}[1, 2, \dots, N] > 2^N$ for all large positive integers N . The prime number theorem implies the corresponding assertion with 2 replaced by any number smaller than e .